Final report: BIE for Learning only on boundaries

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1 Problem and motivation

Many scientific problems are described by **partial differential equations** (PDEs), which can model phenomena such as heat diffusion, and fluid dynamics. Solving these equations numerically is a central task in scientific computing. However, classical numerical methods such as the finite element method (FEM), finite difference method (FDM), or boundary element method (BEM) can become computationally expensive when the geometry of the domain varies or when the solution must be computed repeatedly for many parameter configurations.

Recently, machine learning approaches have been proposed as an alternative way to approximate PDE solutions. In particular, *physics-informed neural networks* (PINNs), that we already discussed in class, because they avoid generating large training datasets. However, PINNs typically require many interior sampling points in the computational domain, which leads to high training costs in high-dimensional problems and potentially unstability through automatic differentiation.

Another difficulty arises when solving PDEs on **unbounded or exterior domains**, which is that we often have to sample the entire domain, which is not practical and not cost-effective, and special treatments are required to enforce the correct behaviour at infinity.

The paper studied in this project proposes a different strategy that combines ideas from physics-informed learning and classical boundary integral formulations. The key observation is that for many linear PDEs, the solution inside the domain can be represented entirely by integrals defined on the *boundary* of the domain. This suggests that learning the solution may not require sampling the full interior of the domain. Instead, the learning problem can be reduced to identifying certain unknown quantities defined only on the boundary. This is very practical because it reduces drastically the number of training point since the boundary has one dimension less than the full domain.

2 Methodology

The proposed method is based on reformulating the PDE boundary value problem using a **boundary integral equation** (BIE). For a large class of linear PDEs, the solution inside the domain can be expressed using Green's functions as an integral over the boundary. This representation introduces an unknown boundary quantity, often referred to as a *boundary potential* or density function. Once this boundary function is known, the solution at any point in the domain can be evaluated directly from the integral representation. Let us take a look at an example for a better understanding:

2.1 Integral formulation with Green's identity

For Pour deux fonctions suffisamment régulières u et v sufficiently differentiable, defined on Ω, we write Green's identity:

$$\int_\Omega (u\Delta v - v\Delta u)\, d\Omega = \int_\Gamma \left(u\frac{\partial v}{\partial n} - v\frac{\partial u}{\partial n} \right) dS,$$

where $\Gamma = \partial\Omega$ is the boundary of the domain and $\frac{\partial}{\partial n}$ is the normal-axisderivative

In order to build an integral representation of the solution, we choose

$$v(x) = G(x, y),$$

where $G(x, y)$ is the fundamental Laplacian function, defined as the solution of

$$\Delta_x G(x, y) = -\delta(x - y),$$

where δ the y-centered Dirac function.

Now suppoer u is harmonic within the full domain (ie $\Delta u = 0$) Then Green's identity becomes

$$u(y) = \int_\Gamma \left(u(x) \frac{\partial G}{\partial n} - G(x, y) \frac{\partial u}{\partial n} \right) dS(x).$$

This equation shows that the value of the solution on an interior point can be expressed as a function of quantities purely defined on the boundary. In dimension 3, we can show that $G(x, y) = \frac{1}{4\pi|x-y|}$, see Appendix for proof.

2.2 Structure of the paper

The issue is then how to know this function on the boundary, because in the general case, the solution is not that simple. So the main idea of the paper is to use a neural operator to approximate this unknown boundary function. Instead of predicting the full solution field in the domain, the model learns a mapping from the boundary geometry and boundary conditions to the boundary potential required by the integral formulation. Because the integral representation already satisfies the PDE in the interior, the learning task only needs to ensure that the boundary conditions are satisfied.

To train the model, points are sampled only along the boundary of the domain. Two sets of boundary points are used: one set for evaluating the integral representation of the solution and another set for enforcing the boundary conditions. The boundary integral is approximated using Monte Carlo sampling. The neural network predicts the boundary potential at the sampled integration points, and the integral approximation produces a predicted value of the solution at the observation points. The training loss measures the discrepancy between these predicted values and the prescribed boundary conditions.

The architecture used to learn this mapping is based on a neural operator framework. The model processes information about the boundary geometry and boundary condition values and outputs the corresponding boundary potential function. This function is then inserted into the boundary integral formula to reconstruct the solution field. Because the PDE is implicitly enforced through the integral representation, the network does not need to compute high-order derivatives during training, which improves numerical stability.

Once the boundary potential has been learned, the solution can be evaluated at any point in the domain, including points outside the domain for exterior problems. This is a key advantage compared to PINNs, which typically require sampling interior points during training.

3 Reproduced Results

We reproduced the two main numerical experiments of the paper: the 2D parametric Laplace equation and the 2D bi-harmonic equation. The 3D Helmholtz scattering experiment was not reproduced, as it requires interfacing with the `BEMpp` library for reference solutions, which was outside the scope of this project.

What we reproduced: The NOMAD architecture (which is a branch MLP with Fourier-feature trunk), the boundary parameterizations from equations (17) and (22) of the paper, the Green's function kernels, and the Monte Carlo boundary integral loss were all implemented from scratch following the paper. The evaluation metric is the relative ℓ^2 error over interior points, consistent with the paper's definition:

$$\text{err} = \sqrt{\frac{\sum_{i=1}^{N} |u_{\text{pred}}(x_i) - u_{\text{true}}(x_i)|^2}{\sum_{i=1}^{N} |u_{\text{true}}(x_i)|^2}}.$$

What we simplified: Due to computational constraints, training was limited to **2 000 steps** for the Laplace model and **1 000 steps** for the bi-harmonic model, compared to the 200 000 steps reported in the paper. The number of Monte Carlo integration points was set to $M = 3\,000$ for Laplace and $M = 1\,000$ for bi-harmonic. All experiments were run on CPU.

3.1 2D Laplace Equation

The model learns to approximate the boundary operator for geometries parameterised by $t \in [1, 2]$, with the boundary condition $u_0(x, y) = e^x \sin(y)$. The training loss decreased from 8.9×10^{-1} at step 1 to approximately 4×10^{-5} at step 2 000, showing consistent convergence (see Figure 1). The relative ℓ^2 errors evaluated at three unseen values of t are reported in Table 1 alongside the paper's reference values.

t	Ours	Paper
1.15	*[3.30%]*	2.85%
1.35	*[3.29%]*	2.87%
1.45	*[3.28%]*	3.00%

Table 1: Relative ℓ^2 errors for the 2D parametric Laplace equation at three values of the geometric parameter t, evaluated on interior points. Training: 2 000 steps (ours) vs. 200 000 steps (paper).

Qualitatively, the predicted field closely matches the ground truth, and the difference plots show small, spatially distributed residuals, consistent with Figure 1 of the paper. The adapted-scale difference plots we produced (not shown in the original paper) reveal that residuals are slightly concentrated near the boundary and in regions of high curvature of Γ_t, while the interior is well-reconstructed.

3.2 2D Bi-harmonic Equation

The bi-harmonic model was trained with a combined loss on both u and $\partial u/\partial n$ on the boundary, following the approach of the paper. The training loss decreased from 8.97 at step 1 to approximately 3×10^{-2} at step 1 000 (see Figure 2). Convergence was noisier than for the Laplace case, which is expected given the higher-order Green's function kernel and the additional normal-derivative term in the loss. We chose certain parameters (eg in learning rates) to make our code run faster, but it significantly made the errors bigger. Results at $t = 1.15$, 1.35, and 1.45 are reported in Table 2.

t	Ours	Paper
1.15	*[16.4%]*	1.08%
1.35	*[16.8%]*	0.90%
1.45	*[18.9%]*	0.77%

Table 2: Relative ℓ^2 errors for the 2D parametric bi-harmonic equation. Training: 1 000 steps (ours) vs. 200 000 steps (paper).

3.3 Training Cost

Each Laplace training run took approximately 5 minutes on CPU. The bi-harmonic run took around 10 minutes. This confirms that the per-step training cost is dominated by the boundary integral computation rather than the neural network forward pass, and that both models remain significantly more expensive per step than a standard PINN on a comparable domain.

4 Critical Analysis

4.1 Accuracy and Convergence Behaviour

For the Laplace equation, the method performs well even with a relatively small number of training iterations. Despite training for only 2 000 steps instead of the 200 000 reported in the paper, the relative ℓ^2 error remains around 3%, which is close to the values reported by the authors. The training curve shows a smooth decrease of the loss over several orders of magnitude, indicating stable optimization.

The situation is significantly different for the bi-harmonic equation. The training loss decreases much more slowly and exhibits large oscillations during optimization. The final relative ℓ^2 errors are between 16% and 19%, which is an order of magnitude larger than the errors reported in the paper. This behaviour can be explained by several factors.

First, the bi-harmonic problem involves higher-order derivatives and more complex Green's function kernels. The boundary integral representation therefore becomes more sensitive to errors in the learned boundary density. Small inaccuracies in the neural operator output can propagate through the integral representation and lead to larger reconstruction errors in the interior of the domain.

Second, the loss function includes both the function value and its normal derivative on the boundary. This makes the optimization problem more difficult and may increase the variance of the Monte Carlo approximation used to estimate the boundary integrals.

4.2 Effect of Monte Carlo Integration

A key component of the method is the Monte Carlo approximation of the boundary integrals. While this allows the method to avoid constructing explicit boundary meshes, it also introduces stochastic noise into the loss function. This effect is visible in the training curves, which display noticeable fluctuations even after the loss has reached small values.

The accuracy of the integral approximation depends directly on the number of integration points M. Increasing M improves the accuracy of the boundary integral but also increases the computational cost of each training step. This creates a trade-off between computational efficiency and numerical accuracy.

In our experiments, the integral computation clearly dominates the training cost. Each training step requires evaluating the Green's function between all observation points and integration points, which scales approximately as $\mathcal{O}(N_y M)$. As a result, the method can become expensive when large numbers of integration points are required for accurate approximations.

4.3 Spatial Distribution of Errors

The difference plots reveal that the largest errors tend to occur near the boundary of the domain and in regions where the geometry has high curvature. This behaviour is consistent with the nature of boundary integral formulations. Since the solution inside the domain is entirely reconstructed from boundary information, any inaccuracies in the learned boundary density can propagate inward.

In contrast, the interior of the domain is typically reconstructed very accurately. This suggests that the neural operator is able to capture the global structure of the solution, while local boundary effects remain harder to approximate.

4.4 Limitations of the Approach

First, the approach relies on the existence of an explicit Green's function for the PDE. This restricts the method primarily to linear PDEs with known fundamental solutions. Extending the approach to nonlinear equations or PDEs with spatially varying coefficients would be significantly more challenging.

Second, the method still requires evaluating singular kernels of the form $G(x, y)$ when x approaches y. In practice, this requires special numerical treatments or clipping strategies to avoid numerical instabilities.

Finally, although the method avoids sampling interior points, it does not eliminate computational cost entirely. The boundary integral evaluation introduces quadratic complexity with respect to the number of boundary samples, which may limit scalability for very complex geometries.

5 Conclusion and Outlook

In this project we studied and implemented the method proposed in *Learning Only on Boundaries: A Physics-Informed Neural Operator for Solving Parametric PDEs in Complex Geometries*. The main idea of the paper is to combine neural operator learning with classical boundary integral formulations in order to avoid sampling the interior of the domain. By expressing the solution of certain linear PDEs as an integral over the boundary, the learning problem can be reduced to approximating an unknown boundary density.

We reproduced two of the main numerical experiments of the paper: the 2D parametric Laplace equation and the 2D bi-harmonic equation. Our implementation confirms that the method can learn accurate solution operators while relying exclusively on boundary samples. In particular, the Laplace experiment achieved relative ℓ^2 errors close to the values reported in the paper, despite using significantly fewer training iterations. The predicted fields also qualitatively match the ground truth solutions across different geometries.

However, our experiments also highlight several limitations of the approach. The bi-harmonic experiment produced significantly larger errors than those reported in the paper, which suggests that higher-order PDEs are more sensitive to inaccuracies in the learned boundary density and to noise in the Monte Carlo approximation of the boundary integral. In addition, the computational cost of each training step is dominated by the evaluation of the boundary integrals, which limits the scalability of the method when the number of boundary points or integration samples becomes large.

Overall, the approach provides a framework that combines classical numerical analysis with modern operator learning techniques. The idea of restricting learning to the boundary is particularly attractive for problems involving complex geometries or unbounded domains, where interior sampling can be expensive or impractical.

Several directions could be explored to improve the robustness and efficiency of the method. First, more accurate quadrature schemes could replace the Monte Carlo approximation in order to reduce variance in the boundary integral computation. Finally, extending the framework to handle nonlinear PDEs or equations with variable coefficients would significantly broaden the applicability of the approach.

6 Appendix

6.1 Fundamental Laplacian function

Let's take a look at the dimension 3. The fundemental Laplacian function is

$$G(x, y) = \frac{1}{4\pi|x - y|}.$$

We obtain this solution by seeking a radial solution of

$$\Delta G = 0$$

for $x \neq y$. Letting $r = |x - y|$, the spherical Laplacian is

$$\Delta g = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dg}{dr} \right).$$

The general solution of this problem is $g(r) = C/r$. With Green-Ostrogradski's theorem, we can conclude that $C = \frac{1}{4\pi}$, but we will not details the steps. Thus, the final solution is

$$G(x, y) = \frac{1}{4\pi|x - y|}.$$

This function describes the influence of a source y, located in the interior, on the solution observed on x. It allows to write the solution of a harmonic equation as the superposition of effects on the boundary.

6.2 Plots

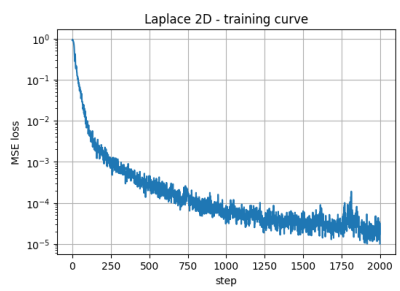

Figure 1: Loss for the Laplace equation

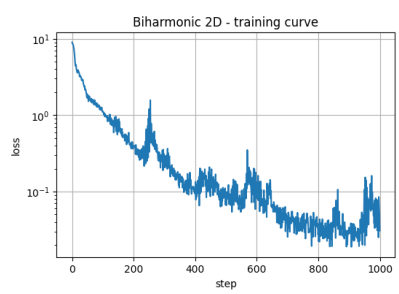

Figure 2: Loss for the Biharmonic equation

7 Notebook

`https://github.com/cedriccm/ML-for-scientific-computing/blob/main/projet-fang-2023.ipynb`